Avinash Kodali

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SUMMARY

Software Engineer with 4+ years of experience developing enterprise applications, distributed data pipelines, and cloud-native services. Built and optimized production systems handling high-volume workloads, with experience spanning Java microservices, full-stack development, and distributed data processing.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL
Backend & Web: Spring (Boot, MVC, Data JPA, Security), Node.js, React, REST APIs, Microservices
Data Engineering: ETL/ELT, Apache Spark, Kafka, Hadoop, Airflow, Apache Hudi, BigQuery
Databases: MySQL, PostgreSQL, Oracle DB, MongoDB, Cassandra
Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, Terraform, Jenkins, CI/CD
AI/ML: scikit-learn, PyTorch, MLOps, RAG, LangGraph

PROFESSIONAL EXPERIENCE

Walmart Global Tech, Bentonville, AR — Software Engineer III — May 2025 – Present

- **Developed scalable batch data pipelines using Apache Spark, Apache Hudi, and Apache Airflow**, migrating high-volume order-processing workloads from Kubernetes-based microservices to distributed processing, contributing to **$1M in projected annual cloud savings** and a **40% reduction in operational alerts.**
- **Optimized 5 production Apache Spark data processing jobs, reducing execution time by 10–30%** through distributed query optimization, hash-based repartitioning, Adaptive Query Execution (AQE) tuning, and shuffle reduction to improve high-volume order-processing performance.
- **Reduced production incidents by 25–40%** through root-cause analysis, targeted reliability improvements, and data-quality checks across Kafka processing, Spark workloads, databases, and application services.
- **Migrated backend services from Java 8/Spring Boot 2.3 to Java 21/Spring Boot 3.x,** resolving compatibility issues and expanding JUnit and integration test coverage to reduce regression risk and improve release reliability.
- Built and deployed an internal **Retrieval-Augmented Generation (RAG) assistant** using embeddings and vector search over operational runbooks and logs; evaluated retrieval quality and tuned chunking and top-k parameters to reduce manual log searches and accelerate incident triage.

Arizona State University, Tempe, AZ — Software Developer — Jun 2024 – May 2025

- **Developed a scalable event-management platform for 1,000+ users** using React and Spring Boot microservices, supporting multi-host event management and stateless services, with Kubernetes deployments on AWS and infrastructure provisioned through Terraform.
- **Reduced event check-in time by 20%** by implementing QR code-based attendee verification, streamlining check-in workflows and improving event entry efficiency.

Arizona State University, Tempe, AZ — Graduate Research Assistant — Jan 2023 – May 2024

- **Developed an end-to-end electricity load-forecasting system** with automated data pipelines and machine learning models, **improving forecast accuracy by 10%** over the previous baseline.
- **Engineered a detection and mitigation framework** for load-redistribution attacks on power grids as part of an Israel–US BIRD Foundation research project, achieving **90% detection accuracy.**

Tata Consultancy Services, India — Systems Engineer — Oct 2020 – Jul 2022

- **Developed a closed-loop Software-in-the-Loop (SiL) simulation platform** for Advanced Driver-Assistance Systems (ADAS), building interactive workflows for **350+ engineers** and **improving testing efficiency by 30%.**
- **Built an automated test-reporting pipeline** to collect simulation results, generate test reports, and streamline failure analysis, reducing manual reporting effort for ADAS engineers.

EDUCATION

M.S. Computer Engineering (Computer Systems) — Aug 2022 – May 2024
Arizona State University, Tempe, AZ

B.Tech. Electrical & Electronics Engineering — Jun 2016 – May 2020
JNTUH College of Engineering, Hyderabad, India

CERTIFICATIONS

Cloud & Infrastructure: Certified Kubernetes Administrator (CKA); HashiCorp Terraform Associate; GCP Associate Cloud Engineer
AWS: Solutions Architect – Associate; Developer – Associate; Cloud Practitioner
Data: Snowflake SnowPro Core